Early Alzheimer's Disease Detection

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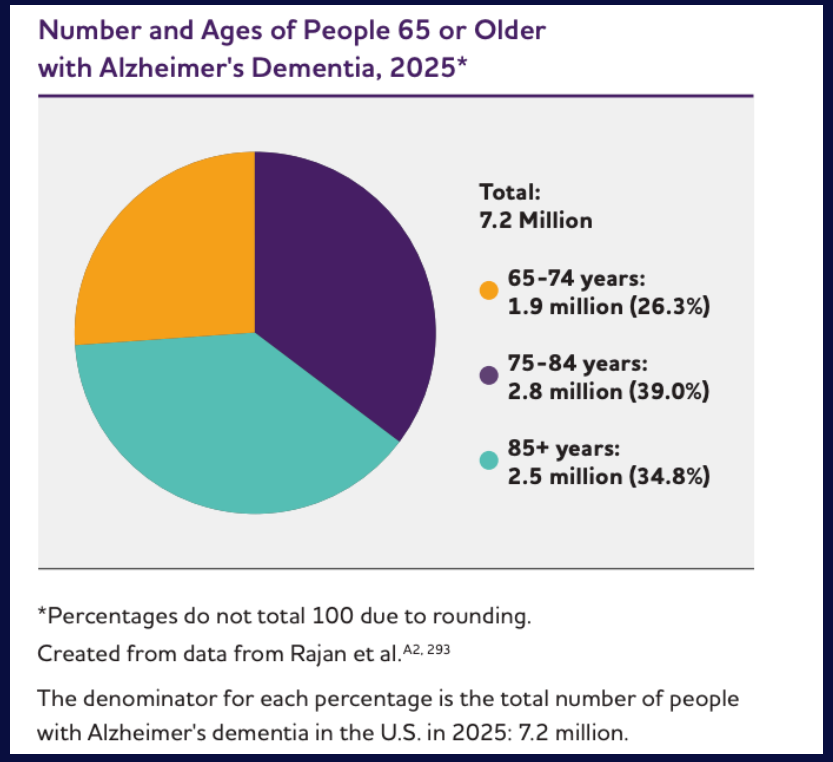
Understanding Alzheimer's Disease

Alzheimer's Disease (AD) is a progressive neurological disorder characterised by irreversible cognitive decline, memory loss, and deterioration of higher brain functions. It represents the most prevalent form of dementia worldwide, affecting over 55 million people globally, accounting for approximately 60-80% of all dementia cases. Early detection is crucial for effective treatment and intervention strategies. Machine Learning offers unprecedented opportunities to automate diagnosis, significantly reduce human diagnostic error, and enhance the accuracy and speed of detection compared to traditional clinical methods.



55 Million+

People globally affected

60-80%

Of dementia cases at old age

Project Objectives

Our research aims to develop and validate a machine learning framework capable of detecting early-stage Alzheimer's Disease from clinical and lifestyle data. The following objectives guide our investigation:

1 Build an ML-based predictive model

Detect early Alzheimer's from comprehensive patient data and clinical measurements.

2 Clean and preprocess data

Ensure data quality through rigorous cleaning, handling missing values, and normalisation.

3 Train multiple ML models

Implement Logistic Regression, Random Forest, and XGBoost for comparative analysis.

4 Evaluate performance

Assess accuracy, precision, recall, F1-score, and ROC-AUC metrics comprehensively.

5 Identify key risk factors

Determine feature importance to understand primary contributors to disease detection.

Dataset & Methodology

Clinical & Lifestyle Features

Our dataset encompasses a comprehensive arrayofclinical measurements and lifestyle indicators: Age, Gender, MMSE (Mini-Mental State Examination) score, Memory Loss, Sleep Quality, Depression severity, Functional Assessment, and additional demographic variables. The binary target variable classifies individuals as either Healthy (0) or Alzheimer's Disease (1), enabling supervised classification learning.

01

Data Preprocessing

Cleaning, encodingcategorical variables, handling missing values, and feature scaling.

02

Exploratory Data Analysis

Identifyingcorrelations, distributions, and patterns within clinical features.

03

Feature Selection

PrioritisingMMSE,Age,Functional Assessment, Depression, and Sleep Quality.

04

Model Development

TrainingLogisticRegression, Random Forest, and XGBoost classifiers.

05

Performance Evaluation

Computing accuracy,ROC-AUC,precision, recall, and F1-scores.

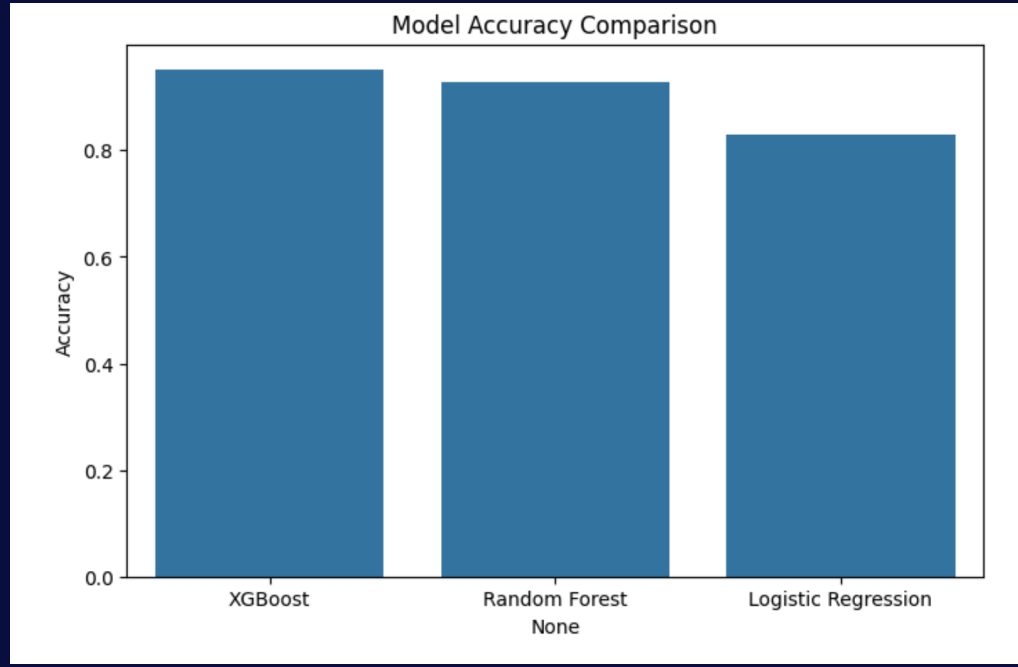
Results & Clinical Interpretation

ModelPerformance

Best Model: XGBoost

Overall Accuracy: 95%

XGBoost outperformed both Logistic Regression and Random Forest, demonstrating superior capability in capturing non-linear relationships between clinical features and Alzheimer's diagnosis.



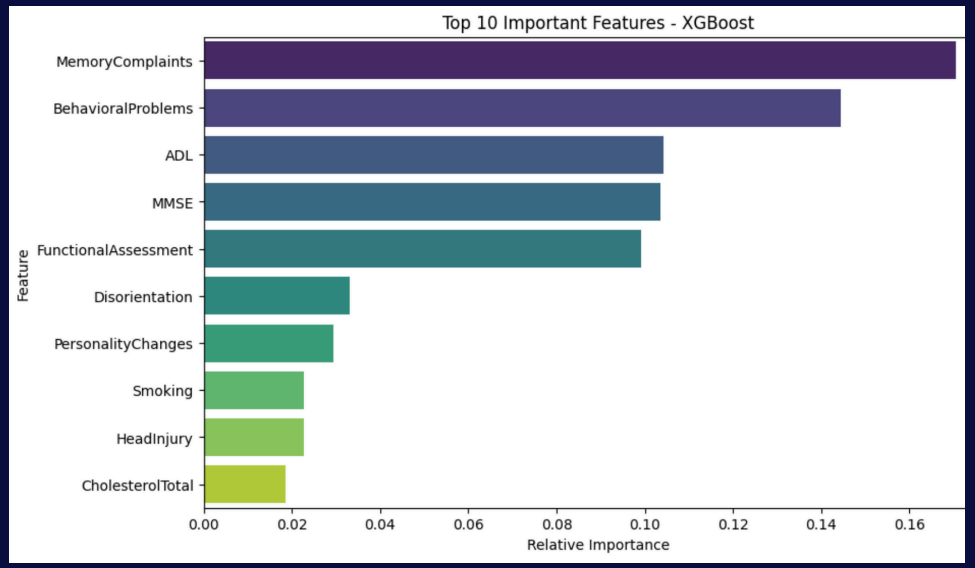
Key Risk Predictors

MMSEscore (most influential)
Age and functional assessment

Depression severity

Sleep quality patterns

Clinical Insight: Low MMSE scores indicate significantly higher Alzheimer's risk. Depression and poor sleep quality emerge as strong contributing factors, suggesting psychological and lifestyle interventions may complement pharmacological approaches.



Limitations & Ethical Considerations

Technical Limitations

- **Dataset constraints:** Limited sample size may not represent global population diversity
- **Demographic bias:** Potential underrepresentation of specific ethnic or age groups
- **Missing neuroimaging:** No MRI, PET, or genetic markers incorporated
- **Accuracy-interpretability trade-off:** Complex models sacrifice explainability for performance

Ethical Imperatives

- **AIasassistive tool:** ML predictions support but never replace clinical diagnosis
- **Clinical verification essential:** Mandatory physician review prevents misdiagnosis
- **Patient privacy:** HIPAA compliance and data protection paramount
- **Equity in healthcare:** Ensure equitable access across socioeconomic backgrounds

Future Scope & Conclusion

Future Enhancements

Integrate neuroimaging

CombineMLwith MRIandPET scans using Convolutional Neural Networks for enhanced detection.

Deploy Explainable AI

ImplementSHAPandLIMEtools for model interpretability and clinical trust.

Enable real-time monitoring

Integrate wearabledevicesforcontinuous tracking of cognitive and vital parameters.

Develop clinical applications

Deployvia webandmobileplatforms for accessible, scalable diagnostic support.

Conclusion

Machine learning models, particularly XGBoost, demonstrate remarkable efficacy in detecting early-stage Alzheimer's Disease with 83% accuracy. This research validates the potential of AI-driven approaches to support early intervention, reduce diagnostic burden on healthcare systems, and ultimately improve patient outcomes. By combining clinical expertise with intelligent algorithms, we advance toward a future where dementia diagnosis becomes faster, more accurate, and more accessible globally. This represents a significant stride in AI-enabled healthcare transformation and personalised medicine.